

Package ‘DDESONN’

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Type Package

Title A Deep Dynamic Experimental Self-Organizing Neural Network Framework

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Description Provides a fully native R deep learning framework for constructing, training, evaluating, and inspecting Deep Dynamic Ensemble Self Organizing Neural Networks at research scale. The core engine is an object oriented R6 implementation with explicit control over layer layout, dimensional flow, forward propagation, backpropagation, and optimizer state updates. The framework does not rely on external deep learning backends, enabling transparent inspection of model state, reproducible numerical behavior, and fine-grained architectural control without requiring compiled dependencies or GPU-specific runtimes. Users can define dimension agnostic single layer or deep multi layer networks without hard coded architecture limits, then run reproducible workflows through high level helpers for fit, run, and predict across binary classification, multiclass classification, and regression modes. Training pipelines support optional self organization, adaptive learning rate behavior, and dynamic ensemble orchestration in which candidate models are evaluated under user specified performance metrics and selectively promoted to a primary ensemble, enabling structured model comparison and controlled ensemble evolution. The framework supports multiple optimization approaches, including but not limited to SGD, RMSProp, Adam, and Lookahead, alongside configurable regularization controls such as L1, L2, and mixed penalties with separate weight and bias update blocks. Activation functions and their corresponding derivatives can be specified per model configuration. Evaluation features provide threshold tuning, relevance scoring, ROC and Precision Recall curve generation, AUC and AUPRC computation, regression error diagnostics, and report ready metric outputs. The package also includes artifact path management, debug state utilities, reproducible scripts, and vignettes that document end to end experiments.

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DDESONN	<i>DDESONN: Deep Dynamic Experimental Self-Organizing Neural Network Framework</i>
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Description

High-level helpers for constructing, training, and evaluating Deep Dynamic Experimental Self-Organizing Neural Networks.

ddesonn_activations *Activation functions (DDESONN)*

Description

A collection of activation functions used by DDESONN. These functions operate on numeric vectors/matrices and preserve shape.

Usage

```
binary_activation(x)

custom_binary_activation(x, threshold = -1.08)

custom_activation(z)

bent_identity(x)

relu(x)

softplus(x)

leaky_relu(x, alpha = 0.01)

elu(x, alpha = 1)

tanh(x)

sigmoid(x)

hard_sigmoid(x)

swish(x)

sigmoid_binary(x)

gelu(x)

selu(x, lambda = 1.0507, alpha = 1.67326)

mish(x)

prelu(x, alpha = 0.01)

softmax(z)

maxout(x, w1 = 0.5, b1 = 1, w2 = -0.5, b2 = 0.5)

bent_relu(x)

bent_sigmoid(x)
```

```

arctangent(x)

sinusoid(x)

gaussian(x)

isrlu(x, alpha = 1)

bent_swish(x)

parametric_bent_relu(x, beta = 1)

leaky_bent(x, alpha = 0.01)

inverse_linear_unit(x)

tanh_relu_hybrid(x)

custom_bent_piecewise(x, threshold = 0.5)

sigmoid_sharp(x, temp = 5)

leaky_selu(x, alpha = 0.01, lambda = 1.0507)

identity(x)

```

Arguments

x	Numeric vector/matrix.
threshold	Threshold for piecewise.
z	Numeric vector/matrix.
alpha	Leak factor.
lambda	SELU lambda.
w1	Weight 1.
b1	Bias 1.
w2	Weight 2.
b2	Bias 2.
beta	Beta parameter.
temp	Temperature/sharpness.

Details

Many functions coerce inputs to matrix form and preserve dimensions. Some functions are experimental and may not be suitable for training.

Value

A matrix of 0/1 values with the same dimensions as x.

A matrix of 0/1 values with the same dimensions as x.

A matrix of 0/1 values with the same dimensions as z .

Bent identity transform.

ReLU transform.

Softplus transform.

Leaky ReLU transform.

ELU transform.

tanh transform.

Sigmoid transform.

Hard sigmoid transform.

Swish transform.

0/1 matrix.

GELU transform.

SELU transform.

Mish transform.

PReLU transform.

Row-wise softmax probabilities.

Elementwise max of two affine transforms.

Bent ReLU transform.

Bent sigmoid transform.

atan transform.

sin transform.

$\exp(-x^2)$.

ISRLU transform.

Bent swish transform.

Parametric bent ReLU transform.

Leaky bent transform.

$x/(1+\text{abs}(x))$.

Hybrid transform.

Piecewise transform.

Sharpened sigmoid.

Leaky SELU transform.

Base identity function.

didesonn_activation_defaults

Default activation sequences for DDESINN helpers

Description

Compute sensible activation functions for hidden and output layers based on the modelling mode and stage (training or prediction).

Usage

```
didesonn_activation_defaults(
  mode = c("binary", "multiclass", "regression"),
  hidden_sizes = NULL,
  stage = c("train", "predict")
)
```

Arguments

mode	Problem mode. One of "binary", "multiclass", or "regression".
hidden_sizes	Integer vector describing the hidden layer widths.
stage	Stage for which activations are required. Either "train" or "predict".

Value

A list of activation functions suitable for passing into the underlying R6 classes.

Examples

```
didesonn_activation_defaults("binary", hidden_sizes = c(32, 16))
didesonn_activation_defaults("regression", hidden_sizes = 64, stage = "predict")
```

didesonn_activation_derivatives

Activation derivatives (DDESINN)

Description

A collection of derivative functions corresponding to DDESINN activation functions.

Usage

```
binary_activation_derivative(x)

custom_binary_activation_derivative(x, threshold = -1.08)

custom_activation_derivative(z)
```

```
bent_identity_derivative(x)
relu_derivative(x)
softplus_derivative(x)
leaky_relu_derivative(x, alpha = 0.01)
elu_derivative(x, alpha = 1)
tanh_derivative(x)
sigmoid_derivative(x)
hard_sigmoid_derivative(x)
swish_derivative(x)
sigmoid_binary_derivative(x)
gelu_derivative(x)
selu_derivative(x, lambda = 1.0507, alpha = 1.67326)
mish_derivative(x)
maxout_derivative(x, w1 = 0.5, b1 = 1, w2 = -0.5, b2 = 0.5)
prelu_derivative(x, alpha = 0.01)
softmax_derivative(x)
bent_relu_derivative(x)
bent_sigmoid_derivative(x)
arctangent_derivative(x)
sinusoid_derivative(x)
gaussian_derivative(x)
isrlu_derivative(x, alpha = 1)
bent_swish_derivative(x)
parametric_bent_relu_derivative(x, beta = 1)
leaky_bent_derivative(x, alpha = 0.01)
inverse_linear_unit_derivative(x)
```

```
tanh_relu_hybrid_derivative(x)

custom_bent_piecewise_derivative(x, threshold = 0.5)

sigmoid_sharp_derivative(x, temp = 5)

leaky_selu_derivative(x, alpha = 0.01, lambda = 1.0507)

identity_derivative(x)
```

Arguments

x	Numeric vector/matrix.
threshold	Threshold.
z	Numeric vector/matrix.
alpha	Leak factor.
lambda	SELU lambda.
w1	Weight 1.
b1	Bias 1.
w2	Weight 2.
b2	Bias 2.
beta	Beta parameter.
temp	Temperature/sharpness.

Value

Vector of zeros.
 Vector of zeros.
 Zero array with same dims.
 Derivative matrix.
 Derivative matrix.
 Derivative matrix.
 Derivative matrix.
 Derivative matrix.
 Derivative matrix.
 Derivative matrix.
 Derivative matrix.
 Derivative matrix.
 Vector of zeros.
 Derivative matrix.
 Derivative matrix.
 Derivative matrix.
 Gradient of active affine branch.
 Derivative matrix.

Derivative matrix (approx).
 Derivative matrix.
 Derivative matrix.
 Derivative matrix.
 Derivative matrix.
 Derivative matrix.
 Derivative matrix.
 Derivative matrix.
 Derivative matrix.
 Derivative matrix.
 Derivative matrix.
 Derivative matrix.
 Derivative matrix.
 Derivative matrix/vector.
 Matrix of ones with same dimensions as x.

ddesonn_artifacts_root

Resolve the writable artifacts root for DDESONN.

Description

Defaults to a user-writable data directory (via `tools::R_user_dir()`) so runtime outputs never write into the installed package tree.

Usage

```
ddesonn_artifacts_root(output_root = NULL)
```

Arguments

output_root	Optional base directory for artifacts. When NULL, a user-scoped directory is selected automatically.
-------------	--

Details

Override order (first non-empty wins):

1. output_root argument
2. `Sys.getenv("DDESONN_ARTIFACTS_ROOT")`
3. `getOption("DDESONN_OUTPUT_ROOT")`

Value

Absolute path to the artifacts directory (created if missing).

ddesonn_debug_state	<i>Inspect internal DDESONN debug state</i>
---------------------	---

Description

Returns a named list snapshot of objects currently stored in the private package debug/state environment.

Usage

```
ddesonn_debug_state()
```

Value

A named list of objects from DDESONN internal state.

ddesonn_dropout_defaults	<i>Default dropout configuration</i>
--------------------------	--------------------------------------

Description

Produce a simple dropout configuration matching the supplied hidden layer sizes.

Usage

```
ddesonn_dropout_defaults(hidden_sizes)
```

Arguments

`hidden_sizes` Integer vector describing the hidden layer widths.

Value

A list of dropout rates for each hidden layer.

Examples

```
ddesonn_dropout_defaults(c(64, 32))
```

ddesonn_fit

*Fit a ddesonn_model with tidy inputs***Description**

Train a `ddesonn_model` (backed by DDESONN) using matrices or data frames, handling label coercion, validation data, and training control defaults.

Usage

```
ddesonn_fit(
  model,
  x,
  y,
  validation = NULL,
  self_org = NULL,
  ...,
  verbose = FALSE,
  verboseLow = FALSE,
  debug = FALSE
)
```

Arguments

<code>model</code>	A model created by <code>ddesonn_model()</code> .
<code>x</code>	Training features.
<code>y</code>	Training targets/labels.
<code>validation</code>	Optional list containing <code>x</code> and <code>y</code> elements for validation.
<code>self_org</code>	Optional logical override for the legacy self-organization phase. TRUE enables <code>self_organize()</code> during training and FALSE disables it. NULL keeps the configured default (<code>self_org = FALSE</code> in <code>ddesonn_training_defaults()</code>). Self-organization acts on input-space topology error (how well neighborhood structure is organized), not on the supervised prediction-loss term.
<code>...</code>	Named overrides for entries in <code>ddesonn_training_defaults()</code> .
<code>verbose</code>	Logical; emit detailed progress output when TRUE.
<code>verboseLow</code>	Logical; emit important progress output when TRUE.
<code>debug</code>	Logical; emit debug diagnostics when TRUE.

Value

The trained model (invisibly). The underlying R6 object is modified in-place and the last training result is stored under `model$last_training`.

See Also

[DDESONN](#)

Examples

```
data <- mtcars
x <- data[, c("disp", "hp", "wt", "qsec", "drat")]
y <- data$am
model <- ddesonn_model(input_size = ncol(x), output_size = 1, hidden_sizes = 8)
ddesonn_fit(model, x, y, num_epochs = 1, lr = 0.05, validation_metrics = FALSE)

# Regression example (mtcars) with explicit scheduler controls.
# If you do NOT want LR decay, set lr_decay_rate = 1.0.
reg_x <- mtcars[, c("disp", "hp", "wt", "qsec", "drat")]
reg_y <- mtcars$mpg
reg_model <- ddesonn_model(
  input_size = ncol(reg_x), # number of input features
  output_size = 1, # one numeric target
  hidden_sizes = c(16, 8), # hidden-layer widths
  classification_mode = "regression" # problem type
)
ddesonn_fit(
  model = reg_model, # model object from ddesonn_model()
  x = reg_x, # training predictors
  y = reg_y, # training target
  num_epochs = 10, # training epochs
  lr = 0.05, # initial learning rate
  lr_decay_rate = 0.5, # decay multiplier (use 1.0 to disable)
  lr_decay_epoch = 20L, # decay step interval in epochs
  lr_min = 1e-5, # lower bound for learning rate
  validation_metrics = FALSE # disable validation metric pass in this example
)
```

ddesonn_model

Create a high-level DDESONN model wrapper

Description

Initialise a `ddesonn_model` (R6) instance backed by the legacy `DDESONN` class, while handling sensible defaults for activations and node counts.

Usage

```
ddesonn_model(
  input_size,
  output_size,
  hidden_sizes = c(64, 32),
  num_networks = 1L,
  lambda = 0.00028,
  classification_mode = c("binary", "multiclass", "regression"),
  ML_NN = TRUE,
  activation_functions = NULL,
  activation_functions_predict = NULL,
  init_method = "he",
  custom_scale = 1,
  N = NULL,
```

```

    ensembles = NULL,
    ensemble_number = 0L,
    verbose = FALSE,
    verboseLow = FALSE,
    debug = FALSE
  )

```

Arguments

<code>input_size</code>	Number of input features.
<code>output_size</code>	Number of outputs.
<code>hidden_sizes</code>	Integer vector describing hidden layer widths.
<code>num_networks</code>	Number of SONN members to initialise within the ensemble.
<code>lambda</code>	Regularisation strength.
<code>classification_mode</code>	Problem mode: "binary", "multiclass", or "regression".
<code>ML_NN</code>	Logical; whether to initialise a multi-layer SONN.
<code>activation_functions</code>	Optional list of activation functions for training.
<code>activation_functions_predict</code>	Optional list of activation functions used during prediction.
<code>init_method</code>	Weight initialisation scheme passed to the legacy constructor.
<code>custom_scale</code>	Optional scaling factor for the initialiser.
<code>N</code>	Optional total node count. If omitted it is inferred from the architecture.
<code>ensembles</code>	Optional pre-existing ensemble container.
<code>ensemble_number</code>	Identifier used when combining multiple ensembles.
<code>verbose</code>	Logical; emit detailed progress output when TRUE.
<code>verboseLow</code>	Logical; emit important progress output when TRUE.
<code>debug</code>	Logical; emit debug diagnostics when TRUE.

Value

A `ddesonn_model` (R6) instance ready for training.

See Also

[DDESONN](#)

Examples

```

model <- ddesonn_model(
  input_size = 5,
  output_size = 1,
  hidden_sizes = c(32, 16),
  classification_mode = "binary"
)

```

`ddesonn_optimizer_options`*Supported optimizer identifiers*

Description

List the optimizer strings understood by the legacy DDESONN training loop.

Usage

```
ddesonn_optimizer_options()
```

Value

A character vector of supported optimiser identifiers.

Examples

```
ddesonn_optimizer_options()
```

`ddesonn_plots_dir`*Resolve the plots directory inside the artifacts root.*

Description

Resolve the plots directory inside the artifacts root.

Usage

```
ddesonn_plots_dir(output_root = NULL)
```

Arguments

<code>output_root</code>	Optional base directory for artifacts. When NULL, a user-scoped directory is selected automatically.
--------------------------	--

Value

Absolute path to the plots directory.

ddesonn_predict

Generate predictions from a fitted ddesonn_model

Description

Internal prediction engine / forward-pass primitive that produces ensemble or per-model predictions from a trained `ddesonn_model`. For user-facing inference, prefer `predict.ddesonn_model()`, which wraps this helper to provide a stable API for type/aggregate/threshold handling and return shapes. Multiclass note: For multiclass classification, `y` should be encoded as integer class indices 1..K (or a one-hot matrix whose columns follow the model's class order), otherwise accuracy comparisons may be incorrect.

Usage

```
ddesonn_predict(
  model,
  new_data,
  aggregate = c("mean", "median", "none"),
  type = c("response", "class"),
  threshold = NULL,
  verbose = FALSE,
  verboseLow = FALSE,
  debug = FALSE
)
```

Arguments

<code>model</code>	A trained model produced by <code>ddesonn_model()</code> .
<code>new_data</code>	New feature matrix or data frame.
<code>aggregate</code>	Aggregation strategy across ensemble members. One of "mean", "median", or "none".
<code>type</code>	Prediction type. "response" returns numeric predictions, while "class" applies thresholding for classification problems.
<code>threshold</code>	Optional threshold override when <code>type = "class"</code> .
<code>verbose</code>	Logical; emit detailed progress output when TRUE.
<code>verboseLow</code>	Logical; emit important progress output when TRUE.
<code>debug</code>	Logical; emit debug diagnostics when TRUE.

Value

A list containing the aggregated prediction matrix and the per-model outputs when `aggregate = "none"`.

See Also

[DDESONN](#)

Examples

```
data <- mtcars
x <- data[, c("disp", "hp", "wt", "qsec", "drat")]
y <- data$am
model <- ddesonn_model(input_size = ncol(x), output_size = 1, hidden_sizes = 8)
ddesonn_fit(model, x, y, num_epochs = 1, lr = 0.05, validation_metrics = FALSE)
preds <- ddesonn_predict(model, x)
head(preds$prediction)
```

ddesonn_run

Run DDESONN across common ensemble scenarios.

Description

This helper re-creates the four orchestration modes that previously lived in TestDDESONN.R:

Usage

```
ddesonn_run(
  x,
  y,
  classification_mode = c("binary", "multiclass", "regression"),
  hidden_sizes = c(64, 32),
  seeds = 1L,
  do_ensemble = FALSE,
  num_networks = if (isTRUE(do_ensemble)) 3L else 1L,
  num_temp_iterations = 0L,
  validation = NULL,
  x_valid = NULL,
  y_valid = NULL,
  model_overrides = list(),
  training_overrides = list(),
  temp_overrides = NULL,
  prediction_data = NULL,
  test = NULL,
  x_test = NULL,
  y_test = NULL,
  prediction_type = c("response", "class"),
  aggregate = c("mean", "median", "none"),
  seed_aggregate = c("mean", "median", "none"),
  threshold = NULL,
  output_root = NULL,
  plot_controls = NULL,
  save_models = TRUE,
  verbose = FALSE,
  verboseLow = FALSE,
  debug = FALSE
)
```


Arguments

x	Training features (the training split) as a data frame, matrix, or tibble.
y	Training labels/targets (the training split).
classification_mode	Overall problem mode. One of "binary", "multiclass", or "regression".
hidden_sizes	Integer vector describing the hidden layer widths.
seeds	Integer vector of seeds. A separate model (or ensemble stack) is trained for each seed.
do_ensemble	Logical flag selecting the ensemble container modes (scenarios C/D). When FALSE, scenarios A/B are executed.
num_networks	Number of ensemble members inside each <code>ddesonn_model()</code> instance.
num_temp_iterations	Number of TEMP iterations to run when <code>do_ensemble = TRUE</code> (scenario D). Ignored otherwise.
validation	Optional validation list with elements x and y.
x_valid	Optional validation features. Overrides <code>validation\$x</code> when set.
y_valid	Optional validation labels. Overrides <code>validation\$y</code> when set.
model_overrides	Named list forwarded to <code>ddesonn_model()</code> allowing custom architectures.
training_overrides	Named list forwarded to <code>ddesonn_fit()</code> for the main run(s). Any argument accepted by <code>ddesonn_fit()</code> may be provided here. Unspecified values fall back to <code>ddesonn_training_defaults()</code> for the given <code>classification_mode</code> and <code>hidden_sizes</code> . See Details and the example showing how to inspect defaults.
temp_overrides	Optional named list forwarded to <code>ddesonn_fit()</code> for TEMP iterations. Defaults to <code>training_overrides</code> . Use this when TEMP candidates should train differently than the main model.
prediction_data	Optional features for prediction. When supplied, predictions are computed for each seed/iteration.
test	Optional test list with elements x and y. When supplied, the final model computes test metrics (loss and, for classification, accuracy) and stores them in <code>result\$test_metrics</code> . The run history (<code>result\$history</code>) mirrors the training metadata (train/validation losses) and appends <code>test_loss</code> when test data is provided.
x_test	Optional test features. Overrides <code>test\$x</code> when set.
y_test	Optional test labels. Overrides <code>test\$y</code> when set.
prediction_type	Passed to <code>ddesonn_predict()</code> .
aggregate	Aggregation strategy within a single model (across ensemble members).
seed_aggregate	Aggregation strategy across seeds. Set to "none" to keep per-seed prediction matrices.
threshold	Optional threshold override for classification prediction.
output_root	Optional directory where legacy-style artifacts are written. When NULL (default) no files are created.

plot_controls	Optional list passed through to <code>ddesonn_fit()</code> as <code>plot_controls</code> . Use this to enable/disable specific report plots or diagnostics (for example, evaluation report settings). The supported structure is defined by <code>ddesonn_fit()</code> ; this function does not create defaults.
save_models	Logical; if TRUE (default) individual models are persisted when <code>output_root</code> is supplied.
verbose	Logical; emit detailed progress output when TRUE.
verboseLow	Logical; emit important progress output when TRUE.
debug	Logical; emit debug diagnostics when TRUE.

Details

- Scenario A – single model (`do_ensemble = FALSE`, `num_networks = 1`).
- Scenario B – single run with multiple members inside a single model (`do_ensemble = FALSE`, `num_networks > 1`).
- Scenario C – main ensemble container (`do_ensemble = TRUE`, `num_temp_iterations = 0`).
- Scenario D – main ensemble plus TEMP iterations (`do_ensemble = TRUE`, `num_temp_iterations > 0`).

The function accepts a training set, optional validation data, and optional prediction features. It repeatedly instantiates `ddesonn_model()` objects, fits them with `ddesonn_fit()`, and (when requested) calls `ddesonn_predict()` to surface aggregated predictions.

Discovering available training overrides

`training_overrides` is a direct pass-through to `ddesonn_fit()`. To see the baseline defaults used by `ddesonn_run()`, call:

```
ddesonn_training_defaults(classification_mode, hidden_sizes)
```

To see all tunable training arguments, see `?ddesonn_fit`.

Value

A list (classed as "ddesonn_run_result") containing the configuration, per-seed models, and optional prediction summaries.

Examples

```
# =====
# DDESINN - FULL example using package data in inst/extdata
# (binary classification; train/valid/test split; scale train-only)
# =====

library(DDESINN)

set.seed(111)

# -----
# 1) Locate package extdata folder (robust across check/install)
# -----
ext_dir <- system.file("extdata", package = "DDESINN")
if (!nzchar(ext_dir)) {
  stop("Could not find DDESINN extdata folder. Is the package installed?",
       call. = FALSE)
}
```

```

# -----
# 1b) Find CSVs (recursive + check-dir edge cases)
# -----
csvs <- list.files(
  ext_dir,
  pattern = "\\\\.csv$",
  full.names = TRUE,
  recursive = TRUE
)

# Defensive fallback for rare nested layouts
if (!length(csvs)) {
  ext_dir2 <- file.path(ext_dir, "inst", "extdata")
  if (dir.exists(ext_dir2)) {
    csvs <- list.files(
      ext_dir2,
      pattern = "\\\\.csv$",
      full.names = TRUE,
      recursive = TRUE
    )
  }
}

if (!length(csvs)) {
  message(sprintf(
    "No .csv files found under: %s - skipping example.",
    ext_dir
  ))
} else {

  hf_path <- file.path(ext_dir, "heart_failure_clinical_records.csv")
  data_path <- if (file.exists(hf_path)) hf_path else csvs[[1]]

  cat("[extdata] using:", data_path, "\n")

# -----
# 2) Load data
# -----
df <- read.csv(data_path)

# Prefer DEATH_EVENT if present; otherwise infer a binary target
target_col <- if ("DEATH_EVENT" %in% names(df)) {
  "DEATH_EVENT"
} else {
  cand <- names(df)[vapply(df, function(col) {
    v <- suppressWarnings(as.numeric(col))
    if (all(is.na(v))) return(FALSE)
    u <- unique(v[is.finite(v)])
    length(u) <= 2 && all(sort(u) %in% c(0, 1))
  }, logical(1))]
  if (!length(cand)) {
    stop(
      "Could not infer a binary target column. ",
      "Provide a binary CSV in extdata or rename target to DEATH_EVENT.",
      call. = FALSE
    )
  }
}

```

```

    }
    cand[[1]]
  }

cat("[data] target_col =", target_col, "\\n")

# -----
# 3) Build X and y
# -----
y_all <- matrix(as.integer(df[[target_col]]), ncol = 1)

x_df <- df[, setdiff(names(df), target_col), drop = FALSE]
x_all <- as.matrix(x_df)
storage.mode(x_all) <- "double"

# -----
# 4) Split 70 / 15 / 15
# -----
n <- nrow(x_all)
idx <- sample.int(n)

n_train <- floor(0.70 * n)
n_valid <- floor(0.15 * n)

i_tr <- idx[1:n_train]
i_va <- idx[(n_train + 1):(n_train + n_valid)]
i_te <- idx[(n_train + n_valid + 1):n]

x_train <- x_all[i_tr, , drop = FALSE]
y_train <- y_all[i_tr, , drop = FALSE]

x_valid <- x_all[i_va, , drop = FALSE]
y_valid <- y_all[i_va, , drop = FALSE]

x_test <- x_all[i_te, , drop = FALSE]
y_test <- y_all[i_te, , drop = FALSE]

cat(sprintf("[split] train=%d valid=%d test=%d\\n",
            nrow(x_train), nrow(x_valid), nrow(x_test)))

# -----
# 5) Scale using TRAIN stats only (no leakage)
# -----
x_train_s <- scale(x_train)
ctr <- attr(x_train_s, "scaled:center")
scl <- attr(x_train_s, "scaled:scale")
scl[!is.finite(scl) | scl == 0] <- 1

x_valid_s <- sweep(sweep(x_valid, 2, ctr, "-"), 2, scl, "/")
x_test_s <- sweep(sweep(x_test, 2, ctr, "-"), 2, scl, "/")

mx <- suppressWarnings(max(abs(x_train_s)))
if (!is.finite(mx) || mx == 0) mx <- 1

x_train <- x_train_s / mx
x_valid <- x_valid_s / mx
x_test <- x_test_s / mx

```

```

# -----
# 6) Run DDESONN
# -----
res <- ddesonn_run(
  x = x_train,
  y = y_train,
  classification_mode = "binary",

  hidden_sizes = c(64, 32),
  seeds = 1L,
  do_ensemble = FALSE,

  validation = list(
    x = x_valid,
    y = y_valid
  ),

  test = list(
    x = x_test,
    y = y_test
  ),

  training_overrides = list(
    init_method = "he",
    optimizer = "adagrad",
    lr = 0.125,
    lambda = 0.00028,

    activation_functions = list(relu, relu, sigmoid),
    dropout_rates = list(0.10),
    loss_type = "CrossEntropy",

    validation_metrics = TRUE,
    num_epochs = 360,
    final_summary_decimals = 6L
  ),

  plot_controls = list(
    evaluate_report = list(
      roc_curve = TRUE,
      pr_curve = FALSE
    )
  )
)
}

```

ddesonn_training_defaults

Construct default training controls

Description

Build a list of training hyperparameters that mirror the expectations of the legacy DDESONN training loop.

Usage

```
ddesonn_training_defaults(
  mode = c("binary", "multiclass", "regression"),
  hidden_sizes = NULL
)
```

Arguments

`mode` Problem mode used to determine sensible defaults.

`hidden_sizes` Integer vector describing the hidden layer widths.

Value

A named list that can be modified and supplied to `ddesonn_fit()`.

Examples

```
ddesonn_training_defaults("binary", hidden_sizes = c(32, 16))

# Inspect regression defaults (includes LR decay by default).
cfg_reg <- ddesonn_training_defaults("regression", hidden_sizes = c(16, 8))
cfg_reg$lr
cfg_reg$lr_decay_rate
cfg_reg$lr_decay_epoch
cfg_reg$lr_min

# If you prefer a fixed LR in regression, disable decay explicitly.
cfg_reg$lr_decay_rate <- 1.0
```

predict.ddesonn_model *Predict method for DDESINN models*

Description

This is the canonical user-facing wrapper around `ddesonn_predict()`. It standardizes arguments (type, aggregate, threshold) and normalizes the output structure for inference workflows. Multiclass note: For multiclass classification, `y` should be encoded as integer class indices 1..K (or a one-hot matrix whose columns follow the model's class order), otherwise accuracy comparisons may be incorrect.

Usage

```
## S3 method for class 'ddesonn_model'
predict(
  object,
  newdata,
  ...,
  aggregate = c("mean", "median", "none"),
  type = c("response", "class"),
  threshold = NULL,
  verbose = FALSE,
```

```

    verboseLow = FALSE,
    debug = FALSE
  )

```

Arguments

object	A ddesonn_model (R6) returned by ddesonn_run() / ddesonn_model().
newdata	Matrix/data.frame of predictors.
...	Unused.
aggregate	Aggregation strategy across ensemble members.
type	Prediction type. "response" returns numeric predictions, while "class" returns class labels for classification problems.
threshold	Optional threshold override when type = "class".
verbose	Logical; emit detailed progress output when TRUE.
verboseLow	Logical; emit important progress output when TRUE.
debug	Logical; emit debug diagnostics when TRUE.

```
print.ddesonn_run_result
```

Print a summary of a DDESONN run result

Description

Print a summary of a DDESONN run result

Usage

```
## S3 method for class 'ddesonn_run_result'
print(x, ...)
```

Arguments

x	A ddesonn_run_result object.
...	Unused.

Value

x, invisibly.

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